TZU-HUNG LO

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EDUCATION

National Taiwan University Sep. 2024 – Dec. 2027 (Expected)
B.S. in Mechanical Engineering Taipei, Taiwan

- Overall GPA: 3.5 / 4.3
- Relevant Courses: Automatic Control, Microprocessor Controlled Systems, Flight and Space Systems Integration

National Cheng Kung University Sep. 2023 – Jun. 2024 (Transferred to NTU)
B.S. in Mechanical Engineering Tainan, Taiwan

- Overall GPA: 4.06 / 4.3
- Outstanding Student Award, Fall 2023

RESEARCH EXPERIENCE

FPGA-Accelerated MPC Flight Controller for Indoor UAV Feb. 2026 – Present
Undergraduate Researcher, NASA Lab, Dept. of Electrical Engineering (Supervisor: Prof. Cheng-Wei Chen)

- Migrated low-level flight control from the Zynq PS to PX4 on a Pixhawk 6C, recalibrating the IMU, optical-flow sensor, motors and power wiring.
- Built the ROS 2 link between PX4 and the lab's FPGA-based MPC solver (developed by senior members), and validated the system in indoor flight with OptiTrack motion capture as ground truth.
- Industry project with ELAN Microelectronics: set up a PX4 SITL environment with injected sensor noise and analyzed how IMU sampling rate (125–1000 Hz) affects flight performance, informing flight-controller chip specifications.

Pneumatic Artificial Muscle Leg Device for Stair-Climbing Assistance Sep. 2025 – Jan. 2026
Undergraduate Researcher, HER Lab, Dept. of Mechanical Engineering (Supervisor: Prof. Biing-Chwen Chang)

- Built pneumatic muscles with Arduino pressure control, reaching 6 cm contraction at 250 kPa (2 cm target).

PROJECTS

Vehicle Dynamics Modeling and Parameter Analysis Sep. 2025 – Present
Vehicle Dynamics Division, NTU Racing Team (Formula Student)

- Built a MATLAB lap-time simulator and swept center-of-gravity position to find the optimal CG height range.
- Converted the pushrod suspension into an equivalent heave / roll model in IPG CarMaker.

Search-and-Rescue Drone – 2nd Place, Compassion Tech Innovation Competition 2026
College Charity Track; also Special Award and Most Popular Award

- Led flight testing and the rescue-ball drop control for a drone that maps disaster areas with a heat map.

Automated Guided Forklift Feb. 2026 – Jun. 2026
Final Project, Microcontroller Systems, NTU

- Designed a five-channel TCRT5000 IR line-tracking array and calibrated servo timing for repeatable fork pickup.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

NTU Mechanical Engineering International Compass (NTUMEIC) Sep. 2025 – Present
President

- Lead a nonprofit helping NTU students pursue study abroad, publishing dozens of alumni stories on Instagram and Medium.
- Organized 4 info sessions and CV clinics (40 attendees) and a mentor program pairing 20 alumni with applicants.

SKILLS & LANGUAGES

Programming & Robotics: Python, C/C++, MATLAB, PX4, ROS 2, Gazebo / SITL, OptiTrack
Software: Simulink, IPG CarMaker, Vivado, Vitis, Arduino, SolidWorks, Inventor, AutoCAD
Languages: English – Fluent (TOEFL iBT 5.0, CEFR C1); Mandarin – Native; German – Beginner (A2 in progress)